

Internstudio AI Launchpad Week 4 Assignment

Topics Covered

1. File Read and Write Operations
2. Exception Handling
3. Python Modules and Libraries
4. Object-Oriented Programming (OOP)

Total Questions: 20

Section A: File Read and Write Operations

Question 1: Student Information Storage

Create a program that asks the user to enter:

- Student Name
- Roll Number
- Course

Store this information in a text file named students.txt. After saving, display a success message.

Question 2: Display File Contents

Using the file created in Question 1:

- Open the file in read mode.
 - Display all stored student records.
 - Count and display the total number of lines present in the file.
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Question 3: Append Employee Records

Create a program that:

- Accepts Employee Name and Department from the user.
 - Appends the data to a file named employees.txt.
 - Allows multiple employee records to be added without deleting previous records.
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Question 4: Search for a Word in a File

Create a text file containing several sentences.

Write a program that:

- Reads the file.
 - Accepts a word from the user.
 - Checks whether the word exists in the file.
 - Displays the number of times that word appears.
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Question 5: Create a Simple Notes Manager

Develop a menu-driven program with the following options:

1. Add a Note
2. View All Notes
3. Exit

Store all notes in a file named notes.txt.

Section B: Exception Handling

Question 6: Safe Division Calculator

Write a program that:

- Accepts two numbers from the user.
 - Performs division.
 - Handles division by zero using exception handling.
 - Displays an appropriate error message.
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Question 7: Integer Input Validation

Create a program that:

- Asks the user to enter age.
 - Ensures only numeric values are accepted.
 - Uses try-except to handle invalid input.
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Question 8: File Not Found Handling

Write a program that:

- Tries to open a file named data.txt.
 - If the file does not exist, display a user-friendly message instead of crashing.
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Question 9: Multiple Exception Handling

Create a program that:

- Accepts two numbers.
 - Performs division.
 - Handles:
 - Invalid input (ValueError)
 - Division by zero (ZeroDivisionError)
 - Displays appropriate messages for each exception.
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Question 10: ATM Withdrawal Simulation

Create a program that:

- Stores a balance amount.
 - Accepts a withdrawal amount from the user.
 - Raises an exception if:
 - The amount entered is invalid.
 - The withdrawal amount exceeds the available balance.
 - Displays suitable messages.
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Section C: Python Modules and Libraries

Question 11: Math Operations Using math Module

Write a program that:

- Accepts a number from the user.
- Displays:
 - Square Root
 - Power of 2

- Factorial
 - Use the math module.
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Question 12: Generate Random Numbers

Create a program using the random module that:

- Generates a random number between 1 and 100.
 - Displays five random numbers.
 - Finds the highest generated number.
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Question 13: Simple Dice Simulator

Using the random module:

- Simulate rolling a dice.
 - Display a random number between 1 and 6.
 - Allow the user to roll multiple times.
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Question 14: Current Date and Time Display

Create a program using the datetime module that:

- Displays the current date.
 - Displays the current time.
 - Displays the current day of the week.
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Question 15: Build Your Own Module

Create a custom module named calculator.py containing functions:

- add()
- subtract()
- multiply()
- divide()

Import this module into another Python file and perform all four operations.

Section D: Object-Oriented Programming (OOP)

Question 16: Student Class

Create a class named Student with:

- Name
- Roll Number
- Marks

Include methods to:

- Accept data
- Display data

Create objects for at least three students.

Question 17: Bank Account Management

Create a class named BankAccount with:

- Account Number
- Account Holder Name
- Balance

Include methods:

- Deposit Money
 - Withdraw Money
 - Display Balance
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Question 18: Inheritance Example

Create:

- A parent class Person
- A child class Teacher

Store and display:

- Name
- Age
- Subject

Demonstrate inheritance by accessing parent class properties from the child class.

Question 19: Method Overriding

Create:

- A parent class Vehicle
- A child class Car

Both classes should have a method named `show_details()`.

Override the method in the child class and demonstrate how method overriding works.

Question 20: Library Management System

Create a class named Book with:

- Book ID
- Title
- Author

Include methods to:

- Add Book Details
- Display Book Details
- Check Availability Status

Create multiple book objects and display their information.

Submission Instructions

- Write all programs in Python.
- Add proper comments wherever necessary.
- Use meaningful variable and function names.
- Test all programs before submission.
- Save each question in a separate Python file with appropriate file names.